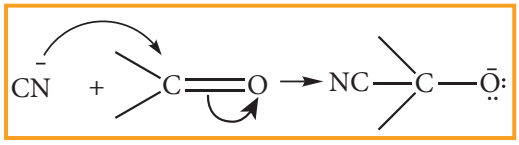
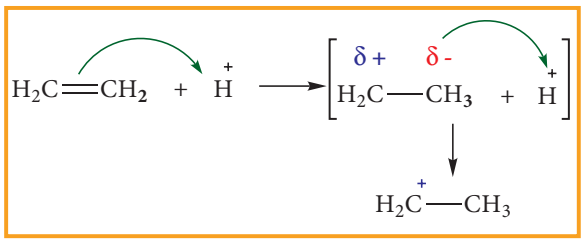
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between the incoming nucleophile and the carbonyl carbon atom.

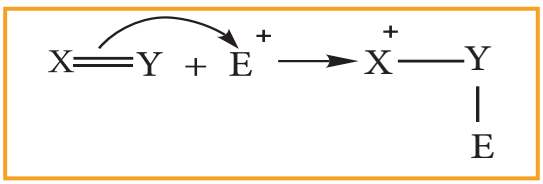


On the other hand when an electrophile such as H^+ approaches an alkene molecule, the **π electrons** are instantaneously shifted to the electrophile and a new bond is formed between carbon and hydrogen. This makes the other carbon electron deficient and hence it acquires a positive charge.



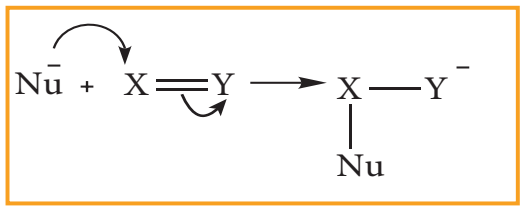
The electromeric effect, is denoted as E effect. Like the inductive effect, the electromeric effect is also classified as +E and -E based on the direction in which the pair of electron is transfered to form a new bond with the attacking agent.

When the π electron is transferred towards the attacking reagent, it is called + E (positive electromeric) effect.



The addition of H^+ to alkene as shown above is an example of +E effect.

When the π electron is transfered away from the attacking reagent, it is called, -E (negative electromeric) effect



The attack of CN^- on a carbonyl carbon, as shown above, is an example of -E effect.

Resonance or Mesomeric effect

The resonance is a chemical phenomenon which is observed in certain organic compounds possessing double bonds at a suitable position. Certain organic compounds can be represented by more than one structure and they differ only in the position of bonding and lone pair of electrons. Such structures are called resonance structures (canonical structures) and this phenomenon is called resonance. This phenomenon is also called mesomerism or mesomeric effect.

For example, the structure of aromatic compounds such as benzene and conjugated systems like 1,3-butadiene cannot be represented by a single structure, and their observed properties can be explained on the basis of a resonance hybrid.

In 1,3 buta diene, it is expected that the bond between C^1-C^2 and C^3 –C^4 should be shorter than that of C^2-C^3, but the observed bond lengths are of same. This property cannot be explained by a simple structure in which two π bonds localised between C^1-C^2 and C^3 –C^4. Actually the π electrons are delocalised as shown below.